

SMA 300: REAL ANALYSIS I

Purpose of the Course

The purpose of this course is to enable learners to develop a deeper and more rigorous understanding of the real number system including defining terms and proving theorems about open and closed sets, sequences, series, limits and continuity of functions.

Course Learning Outcomes

By the end of this course the learner will be able to:

- i) Construct the real number system and apply the concept of completeness in the set of real numbers and its topology.
- ii) Prove and apply theorems associated with sequences of real numbers.
- iii) Classify a given series of real numbers as convergent or divergent.
- iv) Discuss properties of countable and uncountable sets.
- v) Distinguish between point wise and uniform continuity of real-values functions.

Course Content

The Real number System: field and order structure, principle of induction, Rational & Irrational numbers, Bounded sets, Supremum and Infimum, Completeness in the set of real numbers. Topology of the Real Numbers: Open sets, limit points, closed sets and closure of sets. Sequences: Limit point of a sequence. Limit superior and Limit Inferior, Convergent sequences, Monotonic sequence, Cauchy sequence, principle of convergent sequences. Series: Partial sum of series, Convergence of series, Absolute and conditional convergence of series. Tests of convergence: Comparison tests, Cauchy's Root test, D'Alembert's Ratio test, Integral test, Alternating series test. Countable and uncountable sets; Functions: Limits, Continuity and Uniform Continuity.

(Pre-requisite SMA 200: Calculus II)

Core Reading Materials

1. Bartle, R.G., and Sherbert, D.R., Introduction to Real Analysis, 4th Ed., Wiley, 2001.

2. Rudin, W., Principles of Mathematical Analysis, 3rd Ed., McGraw-Hill, 1976.
3. Trench, W.F., Introduction to Real analysis, Pearson Education, 2003.
4. Lewin, J., An Interactive Introduction to Mathematical Analysis, Revised Ed., Cambridge University Press, 2014.

Recommended Reading Materials

1. Lewin, J and Lewin, M., An Introduction to Mathematical Analysis, 2nd Ed. McGraw–Hill, New York, 1993.
2. Nikolsky, S.M., A course of Mathematical Analysis, Vol.2, MIR, Moscow, 1987.

THE REAL NUMBER SYSTEM

Field and Order structure of Real Numbers

Notation

We shall denote by:

$\mathbb{Z}$: the set of all integers

$\mathbb{N}$: the set of all positive integers

$\mathbb{Q}$: the set of all rational numbers

$\mathbb{Q}^c$: the set of all irrational numbers

$\mathbb{R}$: the set of all real numbers

$\forall$: For all (or for every)

$\exists$: There exists (or there is)

$\exists!$: There exists a unique element

$\Rightarrow$: Implies that

$\Leftrightarrow$: If and only if

Algebraic Structure of Real Numbers

Let $+$ denote addition and $\cdot$ denote multiplication. Then

1. For all $x, y \in \mathbb{R}$, we have
 - i) $x + y \in \mathbb{R}$
 - ii) $x \cdot y \in \mathbb{R}$ (Closure)
2. For all $x, y \in \mathbb{R}$, we have
 - i) $x + y = y + x$
 - ii) $x \cdot y = y \cdot x$ (Commutative law)
3. For all $x, y, z \in \mathbb{R}$, we have
 - i) $(x + y) + z = x + (y + z)$
 - ii) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (Associative law)
4. There exist distinct elements 0 and 1 in $\mathbb{R}$ such that
 - $x + 0 = x$
 - $x \cdot 1 = x, \forall x \in \mathbb{R}$Number 0 is called the additive identity
1 is called the multiplicative identity
5. For all $x \in \mathbb{R}$, there exists a real number y such that $x + y = 0$. The number $y \in \mathbb{R}$ is called the additive inverse of x .
Eg if $x = 7$ the $y = -7$.

6. For all $x \in \mathfrak{R}$ such that $x \neq 0$, there exists a real number $y \in \mathfrak{R}$ such that $x \cdot y = 1$.

The number y denoted by x^{-1} is called the multiplicative inverse of x .

Eg if $x = 2$, then $2^{-1} = \frac{1}{2}$.

Definition 1.1: A set with two operations $+$ and $\cdot$ satisfying all the above properties is called a field. Thus the set of all real numbers is a field.

Ordering property of real numbers

Definition 1.2: Let F be a field, an order relation $\leq$ on F is a relation such that

- i) $\forall x, y \in F, x \leq y$ and / or $y \leq x$
 - ii) if $x < y$ and $y < x$ then $x = y$
 - iii) if $x \leq y$ and $y \leq z$ then $x \leq z$
- and if $x \neq y$ we write $x < y$ or $y > x$.

A field with an ordered relation satisfying (i), (ii) and (iii) is called an ordered field.

Example 1.1

The set of all real numbers $\mathfrak{R}$ is an ordered field.

Rational and Irrational Numbers

Rational Numbers

The continuum

We extend the natural numbers by including the number zero and all the negative whole numbers to obtain the set of integers $\{\dots -2, -1, 0, 1, 2, \dots\}$

As long as we are in equation of the type $2x - 5 = 0$ we have no whole number solutions, so we need more numbers to include fractions. The next set is called the set of rational numbers (i.e. we extend $\mathbb{Z}$ to the set of rational numbers)

Definition 1.3: A rational number is a number of the form $r = \frac{m}{n}$, such that $(m, n) = 1$ and, $m, n \in \mathbb{Z}$, $n \neq 0$, that is m and n have no common factor other than 1. The set of rational numbers is denoted by Q , that is $Q = \left\{ \frac{m}{n} : (m, n) = 1, m, n \in \mathbb{Z}, n \neq 0 \right\}$.

The set of rational numbers is closed under both addition and multiplication.

Lemma 1.1

Between any two rational numbers there exists always a third number.

Proof: let v_1 and v_2 be two rational numbers, then $v_3 = \frac{v_1 + v_2}{2}$ is a rational number between v_1 and v_2 .

In fact, this process can be repeated infinitely many times, and so between any two rational numbers, there are infinitely many rationals.

1.2.2.2 Irrational numbers

Definition 1.4: The set of numbers that are not rational cannot be expressed in the form $\frac{m}{n}$ and are called irrational numbers and are denoted by Q^c .

Theorem 1.2: There does not exist a rational number r such that $r^2 = 2$, in other words $r = \sqrt{2}$ is an irrational number.

Proof

Suppose to the contrary that $r = \sqrt{2}$ is rational, then

$$\sqrt{2} = \frac{m}{n} \text{ where } (m, n) = 1, m, n \in \mathbb{Z}.$$

$\Rightarrow 2 = \frac{m^2}{n^2} \Rightarrow m^2 = 2n^2$, we see that m^2 is even. This implies that m is also even (because if $m = 2k - 1$ is odd then its square $m^2 = 2(2k^2 - 2k) + 1$ is also odd). Now let $m = 2k$ for $k \in \mathbb{N}$, then $4k^2 = 2n^2$, so that $n^2 = 2k^2$. Therefore, n^2 is even, and so n is even.

Then both m and n are even, hence have a common factor 2, which is a contradiction. Hence $\sqrt{2}$ is irrational.

Definition 1.5: The set of all rational and irrational numbers is called the set of real numbers or the continuum denoted by $\mathbb{R}$. Thus $\mathbb{R} = Q \cup Q^c$, the set $\mathbb{R}$ is closed under addition and multiplication.

1.2.2.4 Intervals

Let a and b be any two real numbers such that $a < b$. Then

- $(a, b) = \{x : x \in \mathbb{R}, a < x < b\}$ is called an open interval
- $[a, b] = \{x \in \mathbb{R}; a \leq x \leq b\}$ is called a closed interval
- $(a, b] = \{x : x \in \mathbb{R}, a < x \leq b\}$ is called half-open/ half-closed interval.
- $[a, b) = \{x : x \in \mathbb{R} : a \leq x < b\}$ is called half-closed/ half-open interval

Bounded Sets

Definition: Let S be a non-empty subset of $\mathbb{R}$.

- The set S is said to be bounded above if there exists a number $x \in \mathbb{R}$ such that $s \leq x$ for all $s \in S$. Such a number x is called an upper bound of S .
- The set S is said to be bounded below if there exists a number $y \in \mathbb{R}$ such that $y \leq s$ for all $s \in S$. Such a number y is called a lower bound of S .

A set S is said to be bounded if it is both bounded above and bounded below. A set is said to be unbounded if it is not bounded

Example

- i) The set $S = \{x \in \mathbb{R} : x \leq 4\}$ is bounded above. The set has no lower bound.
Note that all real numbers greater than 4 are also upper bound of S.
- ii) The set $S = \{x \in \mathbb{R} : x \geq 2\}$ is bounded below. The set has no upper bound. Note that all real numbers less than 2 are also lower bounds of S.
- iii) Lastly the set $S = \{x \in \mathbb{R} : 2 < x \leq 4\}$ is bounded, with lower bound 2 and upper bound 4.

Definition: Of all the upper bounds of a set S, the least is called a least upper bound or supremum of S denoted by $\sup S$.

That is u is a supremum of S if

- i) u is an upper bound of S, and
- ii) if v is any upper bound of S, then $u \leq v$
- iii) $\forall \epsilon > 0, \exists s \in S : u - \epsilon < s < u$.

Definition (Bounded set): If S is bounded below, then of all the lower bounds, the greatest is called a greater lower bound, or an infimum of S denoted by $\inf S$. That is u is an infimum of S if:

- i) u is a lower of S and
- ii) if v is a lower bound of S, then $u \geq v$
- iii) $\forall \epsilon > 0, \exists s \in S : u < s < u + \epsilon$.

If a supremum of a set belongs to the set, then it is called the maximal element of the set. Similarly if an infimum of a set belongs to the set, then it is called the minimum element of the set.

Example

- a) For the set $S = [-2, 3]$ the
Supremum = maximal element = 3 and the
Infimum = minimum element = -2
- b) For the set $S = (1, 3)$, S has neither maximal or minimum element. The supremum of S is 3 while the infimum is 1.

Completeness Axiom of the set of real numbers $\mathbb{R}$

Every non-empty subset S of $\mathbb{R}$ which is bounded above has a supremum in $\mathbb{R}$ and every non-empty subset S of $\mathbb{R}$ which is bounded below has an infimum in $\mathbb{R}$.

The property of $\mathbb{R}$ as an ordered field taken together with the completeness axiom of $\mathbb{R}$ makes $\mathbb{R}$ a complete ordered field.

Example 1.4

The set of rational numbers does not satisfy the completeness axiom. For consider a non-empty subset S of $\mathbb{Q}$, $S = \{x \in \mathbb{Q} : x^2 < 2\}$. Then S is bounded and $\sup S = \sqrt{2}$, which does not belong to $\mathbb{Q}$. i.e $\sqrt{2} \notin \mathbb{Q}$.

Exercise

Find the supremum, infimum, maximal and minimum elements of the following sets:

a) $S = \{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\}$

b) $S = \left\{\frac{1}{m} + \frac{1}{n} : n, m \in \mathbb{N}\right\}$

c) $S = \left\{\frac{m}{n} : m, n \in \mathbb{N}, m < n\right\}$

d) $S = \left\{1 + \frac{1}{n} : n \in \mathbb{N}\right\}$

e) $S = \left\{(-1)^n \left(1 + \frac{1}{n}\right) : n \in \mathbb{N}\right\}$

Density of Rational Numbers in $\mathbb{R}$

The set of rational numbers is “dense” in $\mathbb{R}$, in the sense that given any two real numbers, there is a rational number between them (Infact, there are infinitely many rational numbers).

Proposition 1.3: If $S = \{\frac{1}{n} : n \in \mathbb{N}\}$ then $\inf S = 0$

Proof-clear

Lemma 1.4: If $t > 0$, $\exists n \in \mathbb{N}$ such that $0 < \frac{1}{n} < t$.

Proof: Since $\inf \{\frac{1}{n} : n \in \mathbb{N}\} = 0$, and $t > 0$, then t is not a lower bound for the set $\{\frac{1}{n} : n \in \mathbb{N}\}$. Thus there exists $n \in \mathbb{N}$ such that $0 < \frac{1}{n} < t$.

Theorem 1.5 (The Density): If x and y are any real numbers with $x < y$ then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

Proof

Assume that $x > 0$. Since $y - x > 0$, it follows from Lemma 1.4 that $\exists n \in \mathbb{N}$ such that $\frac{1}{n} < y - x$. Therefore, we have $nx + 1 < ny$. Since $nx > 0$, $\exists m \in \mathbb{N}$ such that $m - 1 \leq nx < m$ (show this). Therefore $m \leq nx + 1 < ny$ and so $nx < m < ny$. Thus, the rational number $r = \frac{m}{n}$ satisfies $x < r < y$.

Corollary : If x and y are real numbers with $x < y$, then there exists an irrational number such that $x < z < y$.

Proof

By Theorem 1.5 between the real numbers $\frac{x}{\sqrt{2}}$ and $\frac{y}{\sqrt{2}}$, there is a rational number $r \neq 0$

(why?) such that $\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}}$.

Then $z = r\sqrt{2}$ is an irrational number (why?) such that $x < z < y$.

Assessment Questions

Let A and B be any two bounded nonempty subsets of real numbers and let $A+B = \{a+b : a \in A, b \in B\}$. Show that $\sup(A+B) = \sup A + \sup B$.

TOPOLOGY OF REAL NUMBERS

Open sets

Definition 2.1: A subset N of $\mathfrak{R}$ is said to be a neighbourhood of a point $x \in \mathfrak{R}$ if there exists an open interval $(a, b) = (x - \epsilon, x + \epsilon)$ such that $x \in (a, b) \subset N$. We call $(a, b) = (x - \epsilon, x + \epsilon)$ an ϵ -neighbourhood of x for some $\epsilon > 0$.

Example 2.1

1. The set $\mathfrak{R}$ is nbd (neighbourhood) of each of its points
2. The set of rational numbers is not a nbd of each of its points (why?)
3. An open interval (a, b) is a nbd of its points.
4. The empty set ϕ is a nbd of each of its points.

Lemma 2.1

If N_1 and N_2 are nbds of a point $x \in \mathfrak{R}$ then $N_1 \cap N_2$ is also a nbd of x .

Proof (Exercise).

Definition 2.2: Let A be a subset of $\mathfrak{R}$. A point $x \in A$ is called an interior point of A if $\exists$ an open interval (a, b) such that $x \in (a, b) \subset A$. The set of all interior points of a set A is called the interior set denoted by A° or $\text{int } A$.

Example 2.2

1. Let $A = [2, 4]$, a closed interval; then $7/2$ is an interior point of A , but neither 2 nor 4.
2. Note that $[2, 4]^\circ = (2, 4)$ but $(2, 4)^\circ = (2, 4)$
3. $\mathbb{N}^\circ = \phi$, $\mathbb{Z}^\circ = \phi$, $\mathbb{Q}^\circ = \phi$, $\mathfrak{R}^\circ = \mathfrak{R}$

Definition 2.3: A subset B of $\mathfrak{R}$ is open in $\mathfrak{R}$ if for each $x \in B$ there exists a nbd V of x such that $x \in V \subset B$. That is B is open if it contains an open interval about each of its points. We also see that a set B is open if and only if $B^\circ = B$.

Example 2.3

- i) Every open interval is an open set.
- ii) The closed interval is not open
- iii) The empty set is open, since ϕ has no element, so there is no element in ϕ which is not an interior point.
- iv) The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are not open sets.
- v) The set $\mathfrak{R}$ is open since $\mathfrak{R}^\circ = \mathfrak{R}$

The following basic result describes the manner in which open sets relate to the operations of the union and intersection of sets in $\mathfrak{R}$.

Theorem 2.2 (Open sets)

- a) The union of an arbitrary collection of open subsets in $\mathfrak{R}$ is open.
- b) The intersection of any finite collection of open sets in $\mathfrak{R}$ is open.

Proof

a) Let $U = \bigcup_{\alpha \in I} A_\alpha$ where A_α is open for each α . If $U = \emptyset$, then U is open. If $U \neq \emptyset$, let $x \in U$, then $x \in A_\alpha$ for some α . Since A_α is open, there exist an open interval (a, b) such that $x \in (a, b) \subset A_\alpha \Rightarrow x \in (a, b) \subset U$ (since $A_\alpha \subset U \forall \alpha$). Hence U is a nbd of x implying that U is open, since x was arbitrary.

b) Let $(A_i)_{i=1}^n$ be a finite family of open sets. We show that $B = \bigcap_{i=1}^n A_i$ is open.

If $B \neq \emptyset$, then B is open. Let $x \in B$, then $x \in A_i$ for all $i = 1, \dots, n$.

Since A_i is open, then $x \in (x - \epsilon_i, x + \epsilon_i) \subset A_i$ for some $\epsilon_i > 0$ for all i .

Letting $\epsilon = \min\{\epsilon_i\}, i = 1, 2, \dots, n$. Then $x \in (x - \epsilon, x + \epsilon) \subset B$. So

$x \in (x - \epsilon, x + \epsilon) \subset \bigcap A_i = B$. Hence B is open. $\square$

Note that, the intersection of an arbitrary collection of open sets need not be open.

For example:

a) Let $G_n = \left(\frac{-1}{n}, \frac{1}{n}\right)$, for $n \in \mathbb{N}$.

i. e $G_1 = (-1, 1)$, $G_2 = \left(-\frac{1}{2}, \frac{1}{2}\right)$, $G_3 = \left(-\frac{1}{3}, \frac{1}{3}\right)$, ...

Then clearly, G_n is open for each $n \in \mathbb{N}$. However, the intersections.

$G = \bigcap_{n=1}^{\infty} G_n = \{0\}$, which is not open for 0 is not an interior point of $\{0\}$ since $(0 - \epsilon, 0 + \epsilon) \not\subset \{0\}$ for every $\epsilon > 0$.

b) Let $A_n = \left(2 - \frac{1}{n}, 3 + \frac{1}{n}\right)$ for $n \in \mathbb{N}$.

Then A_n is open for each $n \in \mathbb{N}$. However, $\bigcap_{n=1}^{\infty} A_n = [2, 3]$ which is not open.

Exercise

Find the interior of

a) $A = \left\{1 + \frac{1}{10^n} : n \in \mathbb{N}\right\}$

b) $B = \left\{\frac{m}{n} : m, n \in \mathbb{N}, m < n\right\}$

c) $C = \left\{1 + \frac{1}{n} : n \in \mathbb{N}\right\}$

d) $D = \{(n, n+1) : n \in \mathbb{N}\}$

$$e) E = (-\infty, -2) \cup (2, 3) \cup \{4\} \cup [7, \infty)$$

Closed sets

Definition (Limit point): A real number p is a limit point (or an accumulation point or a cluster point) of S if every neighbourhood of p contains at least one point of S different from p . That is if N is the neighbourhood of p , then p is a limit point of S if $(N - \{p\}) \cap S \neq \emptyset$.

The set of all limit points of S is denoted by S' and is called the derived set of S .

Example 2.4

Let $(0,1) \subset [0,1]$. Then every member of $[0,1]$ is a limit point of $(0,1)$. Thus though $\{0,1\}$ do not belong to $(0,1)$ they are its limit points for the nbd $(1-\epsilon, 1+\epsilon)$ of 1 contains infinitely many elements of $(0,1)$.

Note that from this we see that a limit point may or may not belong to the set.

Example

- a) The sets $\mathbb{N}, \mathbb{Z}$ and $\emptyset$ have no limit points.
- b) Every real number is a limit point of the sets $\mathbb{Q}$ and $\mathbb{R}$.

Lemma 2.3

Finite sets have no limit points.

Proof: Consider the set $A = \{x_1, x_2, \dots, x_n\}$. Let p be an arbitrary real number and define $d_1 = |x_1 - p|, \dots, d_n = |x_n - p|$.

Let $r = \min\{d_1, \dots, d_n\}$. Then the nbd $N_p(\frac{r}{2})$ of p contains no point of A . Hence p is not a limit point of A . Since p was arbitrary, A has no limit points.

Next we have a characterization of a limit point of a set.

Theorem 2.4

A real number p is a limit point of a subset S of $\mathbb{R}$ iff every nbd of p contains infinitely many points of S .

Proof

($\Rightarrow$) Let p be a limit point of S and Let N_p be a nbd of p , we show that N_p has infinitely many points. Suppose to the contrary that N_p has a finite number of points of S distinct from p . Then by the above lemma p is not a limit point of S , which is a contradiction? Hence N_p contains infinitely many points of S . Thus every nbd of p has infinitely many of S , since N_p was arbitrary.

($\Leftarrow$) Conversely, given that every nbd of p has infinitely many points of S we have that every nbd of p contains a point of S different from p . Hence p is a limit point of S . $\square$

Definition 2.5 (Closed set): Let X be a set and $S \subset X$. Then S is said to be closed in X if it contains all its limit points. That is, if S is closed and s is a limit point of S then $s \in S$.

Example 2.6

1. The $\emptyset$ is closed for there is no element which is not a limit point of $\emptyset$.
2. The set $[a, b]$ is closed since a, b are limit points of A and $a, b \in A$.
3. Every finite set is closed for there is no limit point that does not belong to the set since finite sets have no limit points.

Theorem 2.5

A set A is closed iff its complement A^c is open.

Proof

Suppose A is closed. We show that A^c is open. If $A^c = \emptyset$, then A^c is open. Suppose that $A^c \neq \emptyset$ and let $x \in A^c$, then $x \notin A$. Since A is closed x is not a limit point of A so there exists a neighbourhood N_x of x such that $N_x \cap A = \emptyset$, which implies that $N \subset A^c$. Thus $x \in N \subset A^c$ implying that A^c is open.

Conversely, suppose that A^c is open. We want to show that A is closed. Let x be a limit point of A . Then every neighbourhood of x is such that $N \cap A \neq \emptyset$, thus $x \notin A^c$ since A^c is open. There $x \in A$ as required. Hence A is closed. $\square$

Theorem 2.6

- a) The intersection of an arbitrary collection of closed sets in $\mathfrak{R}$ is closed.
- b) The union of any finite collection of closed sets in $\mathfrak{R}$ is closed.

Proof

- a) Let $S = \bigcap_{\alpha \in I} A_\alpha$, where each A_α is closed. We will show that S^c is open. Now, by De Morgan's Laws,

$$S^c = \left(\bigcap_{\alpha \in I} A_\alpha \right)^c = \bigcup_{\alpha \in I} A_\alpha^c.$$

Since A_α is closed $\forall \alpha$, A_α^c is open for all α and so $\bigcup_{\alpha \in I} A_\alpha^c = S^c$ is open Theorem

2.2. Hence S is closed.

- b) Similarly, let $S = \bigcup_{i=1}^n A_i$, where each A_i is closed. Now, by De Morgan's Laws,

$$S^c = \left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c.$$

Since A_i is closed $\forall i$, A_i^c is open for all i and since $\bigcap_{i=1}^n A_i^c$ is open Theorem 2.2,

S^c is open. Hence S is closed. $\square$

Remark

The finiteness condition in part b) cannot be removed. For consider the example

$A_n = [\frac{1}{n}, 1]$ which is closed but $\bigcup_{n=1}^{\infty} A_n = (0, 1]$ which is not closed.

Definition (Closure): Let $S \subset \mathfrak{R}$, the set of all limit points of S is denoted by S' and is called the derived set. The closure of S denoted by $\bar{S}$ is defined to be $\bar{S} = S \cup S'$. Note that if S is closed then $\bar{S} = S$.

Definition (Dense set): A subset A of $\mathfrak{R}$ is said to be dense in $\mathfrak{R}$ if $\bar{A} = \mathfrak{R}$. That is every point of X is a limit point of A .

Example

The set of rational numbers is dense in $\mathfrak{R}$.

Proof

For let $x \in \mathfrak{R}$, if $x \in \mathbb{Q}$ we are done, otherwise if $x \notin \mathbb{Q}$ then every neighbourhood N_x of x contains at least one other rational number. Therefore x is a limit point of $\mathbb{Q}$ and so $\mathbb{Q}$ is dense in $\mathfrak{R}$, i.e. $\bar{\mathbb{Q}} = \mathfrak{R}$.

Exercise

1. Find the derived sets of the following sets and state whether they are closed or not:

a) $S = \{\text{all integers}\}$

b) $S = \bigcup \{(n, n+1) : n \in \mathbb{N}\}$

c)

d) $S = \left\{x : x = \frac{1}{n}, n \in \mathbb{N}\right\}$

e) $S = \left\{1 + \frac{(-1)^n}{n} : n \in \mathbb{N}\right\}$

f) $S = \left\{1, -1, 1\frac{1}{2}, -1\frac{1}{2}, 1\frac{1}{3}, -1\frac{1}{3}, 1\frac{1}{4}, -1\frac{1}{4}, \dots\right\}$

2. If A is open and B is closed, prove that $A \setminus B$ is open.

SEQUENCES

Convergence of Sequences

Loosely speaking a sequence is a collection of real numbers obeying a particular pattern or definite rule or order.

Definition: A sequence of real numbers is a function whose domain is the set of natural numbers $N = \{1, 2, \dots\}$ and range is a subset of $\mathbb{R}$. That is $f : N \rightarrow \mathbb{R}$ where for any $n \in N$ there exists a unique number x_n such that $f(n) = x_n$.

We denote a sequence by $\{x_n\}$ or by listing the elements $(x_1, x_2, \dots, x_n, \dots)$ and call x_n the n^{th} term of the sequence.

Example 3.1

a) $x = \left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \dots \right\}$ is the sequence of the even numbers.

b) $x = \left\{ 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots \right\}$
 $= \left\{ \frac{1}{2^n} \right\}$, give the formula of the sequence.

Definition (Bounded sequence): A sequence $\{x_n\}$ is said to be bounded if there is a real number M such that

$$|x_n| \leq M \quad \forall n.$$

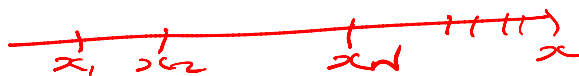
Example

The sequence $s = \left\{ \frac{1}{2^n} \right\}$ is bounded for $|x_n| \leq 1$ or $0 \leq x_n \leq 1$.

But the sequence $s = \{2, 4, 6, 8, \dots\}$
 $= \{2n\}$ is not bounded.

Definition 3.2 (Limit point and convergence of sequences): A sequence $\{x_n\}$ of real numbers is said to converge to x in $\mathbb{R}$ if for every $\epsilon > 0$ there exist a number $N = N(\epsilon)$ such that

$$|x_n - x| < \epsilon, \quad \forall n > N.$$



The number x is called the limit point of the sequence and we write

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{or} \quad x_n \rightarrow x \quad \text{as} \quad n \rightarrow \infty.$$

If the limit of a sequence exists then the sequence is said to be convergent otherwise the sequence is said to be divergent.

Example

Show by first principles that $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right) = 0$.

Proof

We want to show that given $\epsilon > 0$, there is a number $N = N(\epsilon)$ such that

$$|x_n - 0| < \epsilon, \forall n > N.$$

Now,

$$|x_n - 0| = \left|\frac{1}{n} - 0\right| < \epsilon$$

$$\frac{1}{n} < \epsilon$$

$$\Leftrightarrow n > \frac{1}{\epsilon}$$

Certainly $\frac{1}{n} < \epsilon, \forall n > \frac{1}{\epsilon}$.

Therefore, choose N such that $N = \frac{1}{\epsilon}$.

$$(x_n) = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\}$$

$$x_n = \frac{1}{n}$$

$$\left| \frac{1}{n} \right| < \epsilon \Leftrightarrow \frac{1}{n} < \epsilon \quad x_{100}$$

$$\left| \frac{1}{n} \right| = \frac{1}{n} = \frac{1}{100} \text{ eg } \epsilon = \frac{1}{100} \quad x_2$$

$$N = N(\epsilon) = 100$$

$$\epsilon = \frac{1}{2}, N = N(\epsilon) = 2$$

Example 3.4

Show that $\lim_{n \rightarrow \infty} x_n = 1$

where $x_n = 1 + (-1)^n \frac{1}{n^2}$

Proof

Let $\epsilon > 0$ be given. Then

$$|x_n - 1| < \epsilon \Rightarrow$$

$$\left| 1 + (-1)^n \frac{1}{n^2} - 1 \right| < \epsilon$$

$$\left| \frac{(-1)^n}{n^2} \right| < \epsilon$$

$$\Rightarrow n^2 > \frac{1}{\epsilon}$$

$$n > \frac{1}{\sqrt{\epsilon}}$$

Choose N such that, $N = \frac{1}{\sqrt{\epsilon}}$ then

$$|x_n - 1| < \epsilon, \forall n > N.$$

$$|x| = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases}$$

$$(-1)^n, n=1, (-1)^1 = -1$$

$$n=2, (-1)^2 = 1$$

$$n=3, (-1)^3 = -1$$

$$\left| \frac{(-1)^n}{n^2} \right| = \frac{|(-1)^n|}{|n^2|} = \frac{1}{n^2} \quad n \in \mathbb{N}$$

$$\Rightarrow \frac{1}{n^2} < \epsilon$$

Example 3.6

Show that $\lim_{n \rightarrow \infty} \left(\frac{3n+2}{n+1} \right) = 3$.

Proof

Given $\epsilon > 0$ we have,

$$\left| \frac{3n+2}{n+1} - 3 \right| < \epsilon$$

$$\Leftrightarrow \left| \frac{3n+2-3n-3}{n+1} \right| = \left| \frac{-1}{n+1} \right| = \left| \frac{1}{n+1} \right| < \epsilon$$

$$\Leftrightarrow \frac{1}{n+1} < \epsilon$$

$$\Leftrightarrow n+1 > \frac{1}{\epsilon}$$

$$\Leftrightarrow n > \frac{1}{\epsilon} - 1$$

Choose N such that $N = \frac{1}{\epsilon} - 1$, then

$$|x_n - 3| < \epsilon, \quad \forall n > N.$$

$$\frac{|-1|}{|n+1|} = \frac{1}{n+1}$$

Properties of sequences

Theorem 3.1

A sequence $\{x_n\}$ in $\mathbb{R}$ has a unique limit.

Proof

Suppose $\{x_n\}$ has two limits x_1 and x_2 . That is $\lim_{n \rightarrow \infty} x_n = x_1$ and $\lim_{n \rightarrow \infty} x_n = x_2$. Then given

any $\epsilon > 0$, $\exists$ numbers N_1 and N_2 such that,

$$|x_n - x_1| < \frac{\epsilon}{2}, \quad \forall n > N_1$$

and

$$|x_n - x_2| < \frac{\epsilon}{2}, \quad \forall n > N_2.$$

By the triangle inequality for all $n > N = \max\{N_1, N_2\}$,

$$|x_1 - x_2| = |x_1 - x_n + x_n - x_2| \leq |x_1 - x_n| + |x_n - x_2| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ is arbitrary, this implies that

$$|x_1 - x_2| = 0 \Rightarrow x_1 - x_2 = 0 \Rightarrow x_1 = x_2.$$

□

$$N_1, N_2 \rightarrow N \quad x_1 = x_2$$

$$\Rightarrow |x_1 - x_2| = 0$$

$$x_n, x_{n+1} \rightarrow x$$

triangle inequality

$$|x| = \begin{cases} x, & x \geq 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases}$$

$$|x+y| \leq |x| + |y|, \quad \forall x, y \in \mathbb{R}$$

$$\Rightarrow |9+(-7)| = |9-7| = |2| = 2$$

$$|9+(-7)| = 9+7-16 \Rightarrow |9+(-7)| \leq 9+7-16$$

$$|x_1 - x_n| = |x_n - x_1|$$

$$|-(x_n - x_1)| = |-1|(x_n - x_1) = |x_n - x_1|$$

$$\leq |x-y| = |y-x|$$

$$(x_n)_{n=1}^{\infty} \quad (x_n) = M, \quad \forall n \geq 1$$

$$x_n \rightarrow x \quad n \rightarrow \infty$$

Theorem 3.2

Every convergent sequence of real numbers is bounded.

Proof

Suppose that (x_n) converges to x i.e. $\lim_{n \rightarrow \infty} x_n = x$ and let $\epsilon = 1$ be given. Then $\exists$ a

natural number $N = N(\epsilon)$ such that

$$|x_n - x| < 1, \quad \forall n > N.$$

$$\Rightarrow |x_n| = |x_n - x + x| \leq |x_n - x| + |x| < 1 + |x|, \quad \forall n > N$$

Let $M = \max \{|x_1|, |x_2|, \dots, |x_N|, 1 + |x|\}$.

Then $|x_n| \leq M, \quad \forall n$. Hence (x_n) is bounded.

Remark:

Conversely a bounded sequence need not be convergent for instance the sequence

$\{x_n\} = \left\{(-1)^n \left(1 + \frac{1}{n}\right)\right\}$ is bounded since $|x_n| \leq 2, \quad \forall n$; but not convergent since

$\{x_n\} = \left\{-2, \frac{3}{2}, -\frac{4}{3}, \frac{5}{4}, -\frac{6}{5}, \frac{7}{6}, \dots\right\}$ diverges.

Theorem 3.3

Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers such that

$\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} y_n = y$. Then

a) $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y.$

b) $\lim_{n \rightarrow \infty} kx_n = kx, \quad k = \text{constant}.$

c) $\lim_{n \rightarrow \infty} (x_n y_n) = xy.$

d) $\lim_{n \rightarrow \infty} \left(\frac{x_n}{y_n}\right) = \frac{x}{y},$ provided that $y_n \neq 0$ for all n and $y \neq 0$.

Proof

a) To show that $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$, we need to make the difference

$|(x_n + y_n) - (x + y)|$ small for all n sufficiently large.

Using the triangle inequality, we have

$$|(x_n + y_n) - (x + y)| = |x_n - x + y_n - y|$$

$$\leq |x_n - x| + |y_n - y|.$$

Now, given any $\epsilon > 0$, since $x_n \rightarrow x$, there exists N_1 such that $n > N_1$ implies

$|x_n - x| < \frac{\epsilon}{2}$. Similarly, since $y_n \rightarrow y$, there exists N_2 such that $n > N_2$ implies

$|y_n - y| < \frac{\epsilon}{2}$. Thus, if we let $N = \max\{N_1, N_2\}$, then $n > N$ implies

$$\forall \epsilon > 0$$

$$N = N(\epsilon)$$

$$|x_n - x| < \epsilon, \quad \forall n > N$$

$$|x_n| \leq M, \quad \forall n \in \mathbb{N}$$

$$|x + y| \leq |x| + |y|$$

$$x_1, x_2, \dots, x_n, \dots$$

$$n > N$$

$$1/2, 1/4, 1/6, \dots$$

$$1 \text{ or } \rightarrow 0$$

$$|x_n| = \left|(-1)^n \left(1 + \frac{1}{n}\right)\right|$$

$$= \left|(-1)^n\right| \left|1 + \frac{1}{n}\right|$$

$$\leq 1 + \left|\frac{1}{n}\right|$$

$$\leq 2$$

$$-2, -1\frac{1}{2}, -1\frac{1}{3}, \dots$$



Example

$$|kx_n - kx| < \epsilon, \quad \forall n > N$$

$$|x_n - x| < \epsilon, \quad \forall n > N$$

$$x_n \rightarrow x$$

$$n \rightarrow \infty$$

Read

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y.$$

$$\lim (y_n - y) \quad |ab| = |a| |b|$$

b) Exercise

$$x_n y - x_n y$$

c) We use the inequality,

$$+ x_n y_n - x_n y$$

$$\begin{aligned} |x_n y_n - xy| &= |(x_n y_n - x_n y) + (x_n y - xy)| \\ &\leq |x_n y_n - x_n y| + |x_n y - xy| \\ &= |x_n| |y_n - y| + |y| |x_n - x| \end{aligned}$$

$$y (x_n - x)$$

Since the sequence (x_n) converges, it must be bounded. Thus there exists $M_1 > 0$ such that

$$|x_n| \leq M_1, \forall n.$$

Letting $M = \max\{M_1, |y|\}$, we obtain the inequality,

$$|x_n y_n - xy| \leq M |y_n - y| + M |x_n - x|.$$

Now, given any $\varepsilon > 0$, there exists N_1 and N_2 such that

$$|x_n - x| < \frac{\varepsilon}{2M}, \text{ when } n > N_1 \text{ and}$$

$$|y_n - y| < \frac{\varepsilon}{2M}, \text{ when } n > N_2.$$

Now let $N = \max\{N_1, N_2\}$, then

$$if \quad n > N$$

$$|x_n y_n - xy| \leq M |y_n - y| + M |x_n - x| < M \left(\frac{\varepsilon}{2M} \right) + M \left(\frac{\varepsilon}{2M} \right) = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$\Rightarrow x_n y_n \rightarrow xy$, since $\varepsilon > 0$ was arbitrary.

$$\lim_{n \rightarrow \infty} x_n y_n = xy$$

d) Since $\frac{x_n}{y_n} = x_n \left(\frac{1}{y_n} \right)$, it suffices from part c) to show that

$$\lim_{n \rightarrow \infty} \frac{1}{y_n} = \frac{1}{y}.$$

That is give $\varepsilon > 0$ we must make

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \left| \frac{y - y_n}{y_n y} \right| < \varepsilon, \text{ for all } n \text{ sufficiently large.}$$

Since $y \neq 0$, $\exists N_1$, such that $n > N_1 \Rightarrow$

$$|y_n - y| < \frac{|y|}{2}. \text{ Thus for } n > N_1, \text{ we have}$$

$$|y_n| = |y - (y - y_n)| \geq |y| - |y - y_n| > |y| - \frac{|y|}{2} = \frac{|y|}{2}$$

(From $|x - y| > |x| - |y|$)

There also exists N_2 such that $n > N_2$ implies that $|y_n - y| < \frac{1}{2} \in |y|^2$

Let $N = \max \{N_1, N_2\}$. Then $n > N$ implies that

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \left| \frac{y - y_n}{y_n y} \right| < \frac{2}{|y|^2} \cdot \frac{1}{2} |y|^2 = \varepsilon$$

Hence $\lim \frac{1}{y_n} = \frac{1}{y}$.

Thus

$$\lim_{n \rightarrow \infty} \left(\frac{x_n}{y_n} \right) = \frac{x}{y}.$$

□

3.2.3 Subsequences and Cauchy Sequences

Subsequences

Definition 3.3 (Subsequence): Let X be a set and $\{x_n\}$ be a sequence in X . Let (n_k) be a sequence of positive integers such that $n_1 < n_2 < \dots$. Then the sequence (x_{n_k}) is a subsequence of $\{x_n\}$.

Example 3.7

If $\{x_n\} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\}$ then $\{x_{n_k}\} = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots\}$ is a subsequence of $\{x_n\}$.

Theorem 3.4

A sequence $\{x_n\}$ of real numbers converges to x if and only if every subsequence of $\{x_n\}$ converges to x .

Proof

$\Rightarrow$ Suppose $x_n \rightarrow x$. We show that every subsequence (x_{n_k}) of $\{x_n\}$ converges to x .

Thus for every $\varepsilon > 0$ there exists an integer N such that

$$|x_n - x| < \varepsilon, \quad \forall n > N.$$

Take $n_k > N$, then

$$|x_{n_k} - x| < \varepsilon, \quad \forall n_k > N.$$

Since (x_{n_k}) is any subsequence of $\{x_n\}$, it follows that every subsequence of $\{x_n\}$ converges to x whenever $x_n \rightarrow x$

$\Leftarrow$ Conversely, let every subsequence of $\{x_n\}$ converge to x . Then $x_n \rightarrow x$ for $\{x_n\}$ is a subsequence of itself. $\square$

Example 3.8

Consider the sequence defined by

$$x_n = \begin{cases} 1, & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases}$$

That is $\{x_n\} = \{0, 1, 0, 1, 0, \dots\}$. Then the subsequence

$$\{x_{2n}\} = \{1, 1, 1, \dots\} \text{ and converges to } 1.$$

But the subsequence

$$\{x_{2n-1}\} = \{0, 0, 0, \dots\} \text{ and converges to } 0.$$

Since $0 \neq 1$, then $\{x_n\}$ diverges.

3.2.3.2 Cauchy sequences

Definition 3.4: A sequence $\{x_n\}$ of real numbers is said to be a Cauchy sequence if for every $\varepsilon > 0$ there exists an integer $N = N(\varepsilon)$ such that

$$|x_m - x_n| < \varepsilon, \quad \forall m, n > N.$$

Example 3.9

Show that the sequence $\left\{\frac{1}{2^n}\right\}$ is Cauchy.

Proof

We need to show that given $\varepsilon > 0$, there exists an integer $N = N(\varepsilon)$ such that

$$|x_m - x_n| < \varepsilon, \quad \forall m, n > N.$$

Let $m = n + p$, where $p \in \mathbb{N}$, then

$$\Rightarrow |x_m - x_n| = \left| \frac{1}{2^{n+p}} - \frac{1}{2^n} \right| = \frac{1}{2^n} \left(1 - \frac{1}{2^p} \right) < \varepsilon.$$

Since $\left(1 - \frac{1}{2^p}\right) < 1$, then $\frac{1}{2^n} < \varepsilon$

$$\Rightarrow 2^n > \frac{1}{\varepsilon} \Rightarrow n > \frac{\ln\left(\frac{1}{\varepsilon}\right)}{\ln 2}.$$

Choose $N = \frac{\ln\left(\frac{1}{\varepsilon}\right)}{\ln 2}$, hence an integer N exists such that

$$|x_m - x_n| < \varepsilon, \quad \forall m, n > N.$$

Hence $\left\{\frac{1}{2^n}\right\}$ is Cauchy.

Theorem 3.5

Every convergent sequence is Cauchy.

Proof

Suppose $x_n \rightarrow x$. Then for every $\varepsilon > 0$ there exists an integer N such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n > N.$$

Take $m > n$, then we have

$$|x_m - x| < \frac{\varepsilon}{2}, \quad \forall m > n$$

By the triangle inequality,

$$|x_m - x_n| = |x_m - x + x - x_n| \leq |x_m - x| + |x - x_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \forall m, n > N.$$

Thus $\{x_n\}$ is Cauchy. □

Remark 3.1

Not every Cauchy sequence is convergent.

Example 3.10

Let $X = (0, 1] \subset \mathbb{R}$ and consider the sequence $\{x_n\} = \left\{\frac{1}{n}\right\}$ in X . Then $\lim_{n \rightarrow \infty} x_n = 0 \notin X$.

However, $\{x_n\} = \left\{\frac{1}{n}\right\}$ is Cauchy. For let $\varepsilon > 0$ be given, then

$$\left|\frac{1}{n} - 0\right| = \left|\frac{1}{n}\right| = \frac{1}{n} < \varepsilon \Rightarrow n > \frac{1}{\varepsilon}.$$

Hence taking $N = \frac{1}{\varepsilon}$, we have

$$|x_n| < \varepsilon, \quad \forall n > N.$$

Let $m > n$, then

$$|x_m - x_n| < \varepsilon, \quad \forall m, n > N. \text{ Hence } \{x_n\} \text{ is Cauchy.}$$

Monotonic Sequences, Lim sup and Lim inf

Definition 3.5: Let $\{x_n\}$ be a sequence of real numbers, then we say that $\{x_n\}$

- a) Is monotonic increasing, $x_n \uparrow$ if $x_n \leq x_{n+1}$, $\forall n \in \mathbb{N}$.
- b) Is monotonic decreasing, $x_n \downarrow$ if $x_n \geq x_{n+1}$, $\forall n \in \mathbb{N}$.

A sequence $\{x_n\}$ is monotonic if it is increasing or decreasing.

Example 3.10

- i) The sequence $\{1, 2, 4, 8, \dots\} = \{2^n\}$ is monotonic increasing.
- ii) The sequence $\{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots\} = \{\frac{1}{3^n}\}$ is monotonic decreasing.

Theorem 3.5

Let $\{x_n\}$ be a monotonic sequence. Then $\{x_n\}$ is convergent if and only if $\{x_n\}$ is bounded.

Proof

Exercise

Definition 3.6 (Limit Superior, Limit Inferior): Let $\{x_n\}$ be a bounded sequence. A subsequential limit of $\{x_n\}$ is any real number that is the limit of some subsequence of $\{x_n\}$. If S is the set of all subsequential limits of $\{x_n\}$, then we define the limit superior of $\{x_n\}$ to be

$$\text{Limsup } x_n = \overline{\text{Lim } x_n} = \sup S.$$

Similarly, we define the limit inferior of α_n to be

$$\text{Lim inf } x_n = \underline{\text{Lim } x_n} = \inf S.$$

Example 3.11

Let $x_n = (-1)^n + \frac{1}{n}$. We see that

$|x_n| = \left|(-1)^n\right| + \frac{1}{|n|} \leq 2$ for all n , so the sequence $\{x_n\}$ is bounded. The first few terms are

$$0, \frac{3}{2}, -\frac{2}{3}, \frac{5}{4}, -\frac{4}{5}, \frac{7}{6}, -\frac{6}{7}, \dots$$

The subsequence

$\{x_{2n}\} = \{\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \dots\}$ converges to 1 and the subsequence

$\{x_{2n-1}\} = \{0, -\frac{2}{3}, -\frac{4}{5}, -\frac{6}{7}, \dots\}$ converges to -1. Since these are the

subsequential limits, we have

$$\text{Limsup } x_n = \overline{\text{Lim } x_n} = \sup \{-1, 1\} = 1$$

and

$$\liminf x_n = \underline{\lim} x_n = \inf \{-1, 1\} = -1.$$

Theorem 3.6

Let $\{x_n\}$ be a sequence of real numbers, then $\{x_n\}$ converges if and only if $\overline{\lim} x_n = \underline{\lim} x_n$ and the number is real.

Proof

We know that $x_n \rightarrow x$ iff every subsequence of $\{x_n\}$ converges to x . Thus $S = \{x\}$, where S is the set of subsequential limits of $\{x_n\}$. Therefore $\sup S = \inf S = x$, that is

$\overline{\lim} x_n = \underline{\lim} x_n$. Conversely let $\overline{\lim} x_n = \underline{\lim} x_n$, then $S = \{x\}$. Therefore, every subsequence of $\{x_n\}$ converges to x . But $\{x_n\}$ is a subsequence of itself. Hence $\{x_n\}$ converges to x . □

3.3 Exercise

1. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{4n^2 - 3}{5n^2 - 2n} \right) = \frac{4}{5}.$$

2. Prove that $x_n = \frac{n+1}{n+2}$ does **not** converge to 0.

SERIES

Introduction to Infinite Series

Definition 4.1 (Infinite Series): Let (a_n) be an infinite sequence of real numbers. The expression

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

is called an infinite series. The number a_n is called the n^{th} term of the series. If we define

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

•

•

•

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{i=1}^n a_i.$$

Then the sum s_n is called the n^{th} partial sum of the series. The series $\sum_{n=1}^{\infty} a_n$ is said to be convergent if the sequence $\{s_n\}$ of partial sums converges and define

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} s_n.$$

Otherwise the series is said to be divergent.

Example 4.1

The infinite series $\sum_{r=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1.

Solution

We have the partial series given by

$$\begin{aligned} s_n &= \frac{1}{(1)(2)} + \frac{1}{(2)(3)} + \frac{1}{(3)(4)} + \cdots + \frac{1}{n(n+1)} \\ &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \end{aligned}$$

Now $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+1} \right] = 1.$

Some Special Series

Geometric Series

One of the most useful series is the geometric series

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \cdots,$$

where r is called the common ratio.

Now, the n^{th} partial sum

$$s_n = \sum_{k=0}^n ar^k = a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1-r^n)}{1-r}, \quad r \neq 1.$$

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$. Hence the geometric series converges to $\frac{a}{1-r}$ provided $|r| < 1$. If $r \geq 1$, the series diverges.

Example 4.2

Find the sum of the series

a) $\sum_{n=1}^{\infty} \frac{1}{3^{n+1}}$ b) $\sum_{n=0}^{\infty} \frac{2^n}{3}$

Solution

$$\begin{aligned} \text{a) } \sum_{n=1}^{\infty} \frac{1}{3^{n+1}} &= \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \frac{1}{3^5} + \cdots \\ &= \frac{1}{3} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots \right) \\ &= \frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{3^n} \end{aligned}$$

Now,

$\sum_{n=1}^{\infty} \frac{1}{3^n}$ is a geometric series with $r = \frac{1}{3}$ and since $|r| = \frac{1}{3} < 1$

Theorem 4.1 (nth Term Divergence Test)

If $\sum a_n$ is a convergent series, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof

If $\sum a_n$ converges, then the sequence (s_n) of partial sums has a finite limit, say a .

But $a_n = s_n - s_{n-1}$,

so

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = a - a = 0. \quad \square$$

The converse is not true in general, as we shall see later that $\sum 1/n$ diverges (harmonic series) but $\lim_{n \rightarrow \infty} 1/n = 0$.

Remark:

But if for the series $\sum a_n$, $\lim_{n \rightarrow \infty} 1/n \neq 0$, then the series $\sum a_n$ diverges.

Example 4.3

Determine whether the following series converges or diverges;

$$\text{a) } \sum_{n=0}^{\infty} \frac{n}{2n+5} \quad \text{b) } \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

Solution:

a) We use the nth term divergence test

$$a_n = \frac{n}{2n+5} \quad \text{and} \quad \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{2n+5} = \lim_{n \rightarrow \infty} \frac{1}{2 + \frac{5}{n}} = \frac{1}{2} \neq 0$$

$$\text{b) } a_n = \frac{1}{n(n+1)} \quad \text{and} \quad \lim_{n \rightarrow \infty} a_n \frac{1}{n(n+1)} = 0$$

Therefore, the series may or may not converge.

P - series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots,$$

called the harmonic series.

Convergence Tests

The Integral Test:

Theorem 4.2 (Integral test)

Let f be a continuous positive decreasing function of x defined for all $x \geq 1$ by

$f(n) = a_n$. Then the series $\sum a_n$ and the integral $\int_1^{\infty} f(x) dx$ both converges and diverges simultaneously.

Example 4.4

Prove that $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Proof:

Let $f(x) = \frac{1}{x^p}$ and $p > 1$, then f satisfies the hypothesis of the integral test.

Now,

$$\begin{aligned} \text{For } \int_1^{\infty} f(x) dx &= \int_1^{\infty} \frac{1}{x^p} dx = \lim_{n \rightarrow \infty} \int_1^n x^{-p} dx = \lim_{n \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^n \\ &= \lim_{n \rightarrow \infty} \left(\frac{n^{1-p}}{1-p} - \frac{1}{1-p} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n^{1-p}}{1-p} + \frac{1}{p-1} \right) = \frac{1}{p-1} < \infty \end{aligned}$$

(Because $p > 1$, $1-p < 0$).

Since $\frac{1}{p-1}$ is a finite number, then $\sum \frac{1}{n^p}$ converges if $p > 1$ by integral test.

Next, if $p < 1$,

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x^p} dx = \lim_{n \rightarrow \infty$$

Thus $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Example 4.5

1. Apply the integral test on the series $\sum \frac{1}{n^2 + 1}$.

Solution:

Let $f(x) = \frac{1}{x^2 + 1}$, then clearly f satisfies the hypothesis of the integral test.

Now,

$$\begin{aligned} \int_1^{\infty} f(x) dx &= \lim_{n \rightarrow \infty} \int_1^n \frac{1}{x^2 + 1} dx = \lim_{n \rightarrow \infty} [\tan^{-1} x]_1^n \\ &= \lim_{n \rightarrow \infty} [\tan^{-1} n - \tan^{-1} 1] \\ &= \lim_{n \rightarrow \infty} [\tan^{-1} n - \pi/4] \\ &= \pi/2 - \pi/4 = \pi/4 < \infty \end{aligned}$$

Since $\int_1^{\infty} f(x) dx$ converges, then $\sum \frac{1}{n^2 + 1}$ converges by integral test

2. $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges since it is a p -series with $p = 2 > 1$.

Comparison test

Theorem 4.3

Let $\sum a_n$ and $\sum b_n$ be infinite series of non-negative terms. That is, $a_n \geq 0$ and $b_n \geq 0$ for all n . Then

- a) If $\sum a_n$ converges and $0 \leq b_n \leq a_n$ for all n , then $\sum b_n$ converges.
- b) If $\sum a_n$ diverges and $0 \leq a_n \leq b_n$ for all n , then $\sum b_n$ diverges.

Example 4.6

Consider the series

$$\sum \frac{1}{(n+1)^2}. \quad \text{Now for all } n \in \mathbb{N}$$

Since the series $\sum \frac{1}{n(n+1)}$ is convergent, so this implies that the series

$\sum \frac{1}{(n+1)^2}$ also converges.

Example 4.6

Determine whether each of the following series converges or diverges;

a) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$

b) $\sum_{n=1}^{\infty} \frac{1}{2n-1}$

c) $\sum_{n=0}^{\infty} \frac{1}{n!}$

Solution

i) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1} < \sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} < \infty,$

i.e. $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges being a p-series with $p = \frac{3}{2} > 1$.

Hence by comparison test $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$ converges.

ii) $\frac{1}{2n-1} > \frac{1}{2n}$ for all $n \geq 1$.

Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{2n-1} > \sum_{n=1}^{\infty} \frac{1}{2n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} > \infty$$

i.e. $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges being the harmonic series.

Hence $\sum_{n=1}^{\infty} \frac{1}{2n-1}$ diverges by comparison test.

iii) $\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots + \frac{1}{n!} + \dots$

Hence $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges by comparison test.

Absolute Convergence and Conditional Convergence

Absolute convergence

Definition 4.2: Let (a_n) be a sequence in $\mathfrak{R}$. If the series $\sum |a_n|$ converges then the series $\sum a_n$ is to be absolutely convergent in $\mathfrak{R}$.

Theorem 4.4

If a series is absolutely convergent then it is convergent.

Proof

Since $a_n \leq |a_n|$, for all n we have

$$\sum a_n \leq \sum |a_n| < \infty$$

Hence $\sum a_n$ converges by Comparison Test. □

Corollary 4.7

If $\sum a_n$ diverges, then $\sum |a_n|$ also diverges.

Example 4.7

The series $\sum (-1)^n \frac{1}{n^2}$ converges absolutely for $\left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2}$.

Thus $\sum \left| \frac{(-1)^n}{n^2} \right| = \sum \frac{1}{n^2}$ which is converges being a p -series with $p = 2$.

Hence the series $\sum \frac{(-1)^n}{n^2}$ converges absolutely.

The Alternating Series

Definition (Alternating Series) 4.3: .Alternating series is a series in which terms alternate between positive and negative values;

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots.$$

If (a_n) is a decreasing sequence of positive numbers and $\lim_{n \rightarrow \infty} a_n = 0$, then the series $\sum (-1)^{n+1} a_n$ converges.

Example 4.8

The series $\sum (-1)^{n+1} \frac{1}{n}$ is convergent.

Solution:

Since the sequence $\left(\frac{1}{n}\right)$ is decreasing and $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right) = 0$, we have that the alternating series $\sum (-1)^{n+1} \frac{1}{n}$ converges.

Example 4.9

The series $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^n}$ converges since the sequence $\left(\frac{1}{n^n}\right)$ is decreasing and has $\lim_{n \rightarrow \infty} \left(\frac{1}{n^n}\right) = 0$ (by L' hospital's Rule).

Conditional Convergence

Definition 4.4 (Conditional Convergence): A convergent sequence that is not absolutely convergent is said to be conditionally convergent. That is if $\sum a_n$ converges but $\sum |a_n|$ diverges, then $\sum a_n$ is said to be conditionally convergent.

Example 4.10

Determine the conditional convergence of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$.

Solution:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$$

Since the sequence $\left(\frac{1}{\sqrt{n}}\right)$ is decreasing and $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}}\right) = 0$, we have that the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges.

Now,

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{\sqrt{n}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}} >$$

That is $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ diverges being a p -series with $p = 1/2 < 1$.

Hence $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ is conditionally convergent.

The Ratio and Root Test

Theorem 4.4 (Ratio Test)

Let $\sum a_n$ be a series of nonzero terms

- a) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, then the series converges absolutely.
- b) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$, then the series diverges.
- c) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, then the test provides no information.

Example 4.11

Determines whether or not the series $\sum \frac{n^2}{2^n}$ converges or diverges.

Solution

$$a_n = n^2/2^n, \quad a_{n+1} = \frac{(n+1)^2}{2^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{2} \left(\frac{n+1}{n} \right)^2 \right| = \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 + \frac{1}{n} \right)^2 = \frac{1}{2} < 1.$$

Hence the series converges by Ratio Test.

Example 4.12

Determine the convergence or divergence of the series $\sum 1/n^2$.

Solution

$$a_n = 1/n^2, \quad a_{n+1} = \frac{1}{(n+1)^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1} \right)^2 \right| = \lim_{n \rightarrow \infty} \left$$

Determine the convergence or divergence of the series $\sum \frac{n!}{n^n}$.

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)n^n}{(n+1)^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n = \frac{1}{e} < \infty \text{ (L' Hospital's Rule)}\end{aligned}$$

Hence the series converges by Ratio Test.

Note: The test is usually effective with series involving factorial expressions.

Example 4.13

Determine the convergence or divergence of the series $\sum 3^n e^{-n}$.

Solution

$$a_n = 3^n e^{-n}, \quad a_{n+1} = 3^{n+1} e^{-(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{3^{n+1}}{e^{n+1}} \cdot \frac{e^n}{3^n} \right| = \lim_{n \rightarrow \infty} \frac{3}{e} = \frac{3}{e} > 1$$

Hence the series diverges by Ratio Test.

Theorem 4.5 (Root Test)

Given a series $\sum a_n$ with all terms positive, and

$$\text{let } \rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_n}$$

- a) If $\rho < 1$ then the series converges absolutely.
- b) If $\rho > 1$, then the series diverges.
- c) If $\rho = 1$, then the test gives no information.

Example 4.14

The series $\sum 2^n e^{-n}$ converges.

Solution:

$$\text{For } \lim_{n \rightarrow \infty} \left(\frac{2^n}{e^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{2}{e} = \frac{2}{e} < 1.$$

Hence the series converges by Root Test.

Example 4.15

The series $\sum 3^n e^{-n}$ diverges.

For

$$\lim_{n \rightarrow \infty} \left(\frac{3^n}{e^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left| \frac{3}{e} \right| = \frac{3}{e} > 1.$$

Exercise

1. Determine whether the following series converge or diverge;

a) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$

b) $\sum \frac{(-3)^n}{n!}$

c) $\sum \frac{(-5)^n}{2^n}$

2. Determine whether the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n^2-1}}$ converges absolutely, conditionally or not at all.

a) $\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n^2-1}}$

b) $\sum \frac{(-1)^n}{n+3}$

3. Apply the integral test to the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$.

4. Determine the convergence or divergence of each of the following series. Give reasons in each case:

a) $\sum n^3 / 2^n$

b) $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$

c) $\sum \frac{\sin^2 n}{n^2}$

d) $\sum \frac{(-1)^{n+1} n}{n^2+1}$

$$\text{e) } \sum \frac{n!}{n^n}$$

COUNTABLE AND UNCOUNTABLE SETS

Countable sets

Definition 5.1 (Infinite set): A non-empty subset S of $\mathfrak{R}$ is said to be infinite if for any given natural number n , however large there is a subset of S containing more than n elements.

Example 5.1

The set $S = \{x \in \mathfrak{R} : x > 2\}$ is infinite.

Definition 5.2 (Countable set): A set S is said to be countable if it is finite or if it is infinite and its elements can be put into 1-1 correspondence with the set of natural numbers. That is S can be listed as infinite sequence $\{x_1, x_2, x_3, \dots\}$

Example 5.2

The set of all even (odd) numbers is countable for its elements can be put into 1-1 correspondence with the set of natural numbers by definition. That is, there exists a function $f : \mathbb{N} \rightarrow \mathbb{E}$ defined by

$f(n) = 2n$ for even numbers and $f(n) = 2n-1$ for odd numbers that sets up a 1-1 correspondence.

Example 5.3

The set $\mathbb{Z}$ of all integers is countable.

Proof

Consider the following arrangement of the sets $\mathbb{Z}$ and $\mathbb{N}$;

$\mathbb{Z} : 0 \ 1 -1 \ 2 -2 \ 3 -3 \dots$

$\mathbb{N} : 1 \ 2 \ 3 \ 4 \ 3 \ 4 \ 5 \dots$

Then we can define a function $f : \mathbb{N} \rightarrow \mathbb{Z}$ which sets up a 1-1 correspondence as follows:

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is even} \\ -\frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}.$$

Properties of countable sets

Theorem 5.1

A subset of a countable set is countable.

Proof:

Let A be a subset of a countable set B . By definition of a countable set, there exists a 1-1 function $f : B \rightarrow \mathbb{N}$. Now, restrict to the domain of f to A to give a new function

$g : B \rightarrow N$. Function g is 1-1 too since f is. So g is a 1-1 function that maps A into N . Thus A is a countable set. $\square$

Theorem 5.2

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of countable sets. We show that $\bigcup_{n=1}^{\infty} A_n$ is countable.

Proof

For each A_n there exists 1-1 correspondence. $f_n : N \rightarrow A_n$, so we can write

$A_n = \{A_{n1}, A_{n2}, \dots\}$ where $f_n(j) = A_{nj}$. We can now arrange the elements of $\bigcup_{n=1}^{\infty} A_n$ in a rectangular array:

$A_1 : A_{11} \rightarrow A_{12} A_{13} \rightarrow A_{14} \dots$
 $A_2 : A_{21} \swarrow A_{22} \nearrow A_{23} \swarrow A_{24} \dots$
 $A_3 : A_{31} \searrow A_{32} \swarrow A_{33} A_{34} \dots$
 $A_4 : A_{41} \swarrow A_{42} A_{43} A_{44} \dots$
 $\vdots$

By moving along each diagonal of the array in the manner indicated, we obtain a listing of all elements in $\bigcup_{n=1}^{\infty} A_n$;

$\{A_{11}, A_{12}, A_{21}, A_{31}, A_{22}, A_{13}, A_{14}, A_{23}, A_{32}, A_{41}, \dots\}$.

This procedure results in an infinite sequence of elements $\{z_1, z_2, z_3, \dots\}$ of A which can be put into 1-1 correspondence with the set of natural numbers. Hence $\bigcup_{n=1}^{\infty} A_n$ is countable. $\square$

Uncountable set

Definition 5.3 Uncountable sets): A set S is said to be uncountable if it is not countable. That is if there is no one-one correspondence (function) between the elements of S and the set of natural numbers.

Example 5.4

The set $\mathfrak{R}$ of real numbers is uncountable.

Proof

Suppose that $\mathfrak{R}$ countable. Then $\mathfrak{R}$ can be written as a sequence of numbers $\mathfrak{R} = \{x_1, x_2, x_3, \dots\}$. Choose $a_1, b_1 \in \mathfrak{R}$ such that $x_1 \notin [a_1, b_1]$. Having done this choose a_2, b_2 such that $a_1 \leq a_2 < b_2 \leq b_1$ and $x_2 \notin [a_2, b_2]$. Continuing this way, choose a_{n+1}, b_{n+1} such that $a_n \leq a_{n+1} < b_{n+1} \leq b_n$ and $x_{n+1} \notin [a_{n+1}, b_{n+1}]$.

Note that for each n , $x_n \notin [a_n, b_n]$ and the numbers $x_1, x_2, \dots, x_n$ cannot be in the interval $[a_n, b_n]$.

Define $A = \{a_n \mid n \in \mathbb{Z}^+\}$ and $\alpha = \sup A$. Clearly α is a real number and so $\alpha = x_N$ for some natural number N . Since α is an upper bound of A , $a_N \leq \alpha$ and since b_N is an upper bound of A and α is the least upper bound of A , we see that $\alpha \leq b_N$. We therefore conclude that $x_N \in [a_N, b_N]$, contradicting the choice of a_N and b_N which shows that the set $\mathfrak{R}$ is uncountable.

Exercise

1. If the set F is countable and E is non-empty subset of F . Show that E is countable.
2. If the set E is countable, show that the set $E \cup \{a\}$ is countable.
3. Show that the set $\mathbb{Q}^c$ of irrational numbers uncountable.

FUNCTIONS

Limits of functions

Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}$$

The function is defined for all x except at $x = 2$. If $x \neq 2$, then $f(x) = x + 2$.

Investigate what happens to the function $f(x)$ as $x \rightarrow 2$.

It can be shown that as $x \rightarrow 2$, $f(x) \rightarrow 4$.

In this case 4 is called the limit value of $f(x)$ as $x \rightarrow 2$ written:

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right) = 4.$$

Definition 6.1 (Limit): Let $S \subseteq \mathfrak{R}$ and $f : S \rightarrow \mathfrak{R}$ be a function. A real number L is said to be a limit point of f at point $a \in S$ if given any $\varepsilon > 0$, $\exists \delta > 0$ such that if $x \in S$ $|f(x) - L| < \varepsilon$ whenever $|x - a| < \delta$.

Note that f may or may not be defined as $x = a$.

We say that $f(x) \rightarrow L$ as $x \rightarrow a$ and write

$$\lim_{x \rightarrow a} f(x) = L.$$

Example 6.1

Let $f(x) = x^2 + 1$, $\forall x \in \mathfrak{R}$. Prove that $\lim_{x \rightarrow 2} f(x) = 5$.

Solution

We need to show that given $\varepsilon > 0$, $\exists \delta > 0$ such that

$$|x - 2| < \delta \Rightarrow |f(x) - 5| < \varepsilon.$$

Now for $x \in \mathfrak{R}$,

$$|f(x) - 5| = |x^2 + 1 - 5| = |x^2 - 4| = |x - 2||x + 2| < \delta|x + 2|$$

Choose $\delta = \frac{\varepsilon}{|x + 2|}$. Then whenever $|x - 2| < \delta$ we have

$$|f(x) - 5| = |x - 2||x + 2| \cdot \frac{\varepsilon}{|x + 2|} = \varepsilon.$$

Hence $\lim_{x \rightarrow 2} f(x) = 5$.

Properties of limits

(Proofs are similar to those as sequences)

- a) If $\lim_{x \rightarrow a} f(x)$ exists then it is unique.
- b) If $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} g(x) = L_2$, then
 - i) $\lim_{x \rightarrow a} [f(x) + g(x)] = L_1 + L_2$
 - ii) $\lim_{x \rightarrow a} [f(x)g(x)] = L_1L_2$
 - iii) $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{L_1}{L_2}$ provided $g(x) \neq 0 \ \forall x$ and $L_2 \neq 0$.

One-sided limits

Definition 6.2: Let $S \subseteq \mathfrak{R}$ and $f : S \rightarrow \mathfrak{R}$ be a function. If for every $x \in S$ $f(x) \rightarrow L$ as $x \rightarrow a$ and $x > a$ always, then we say that $x \rightarrow a$ from the right and write $x \rightarrow a^+$ and say that

$$\lim_{x \rightarrow a^+} f(x) = L.$$

Similarly,

If $f(x) \rightarrow L$ as $x \rightarrow a$ and $x < a$ always, then we say that $x \rightarrow a$ from the left and write $x \rightarrow a^-$ and say that

$$\lim_{x \rightarrow a^-} f(x) = L.$$

The limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ are called one-sided limits of f at a .

Theorem 6.1

Let $S \subseteq \mathfrak{R}$. Then

$$\lim_{x \rightarrow a} f(x) = L \quad \text{iff} \quad \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L.$$

Example 6.2

- a) $\lim_{x \rightarrow 0} \frac{x}{|x|}$ does not exist

For

$$\lim_{x \rightarrow 0^+} \frac{x}{|x|} = 1$$

and

$$\lim_{x \rightarrow 0^-} \frac{x}{|x|} = -1$$

- b) $\lim_{x \rightarrow 0} \frac{1}{x^2}$ and $\lim_{x \rightarrow 0} \frac{1}{x}$ have infinite limit.

c) $\lim_{x \rightarrow \infty} \frac{\sqrt{x-x}}{\sqrt{x+x}}$ has infinite limit.

Exercise

- Show that from definition of a limit:

a) $\lim_{x \rightarrow 1} \frac{x^2 - x + 1}{x + 1} = \frac{1}{2}$

b) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - x - 6} = \frac{12}{5}$

- Find

a) $\lim_{x \rightarrow 0^+} \frac{4x}{|x|}$

b) $\lim_{x \rightarrow 0^-} \frac{4x}{|x|}$

c) $\lim_{x \rightarrow 1^-} \frac{x^2 - 1}{|x - 1|}$

Continuity of a function

Definition 6.3 (Continuity): A function $f : S \rightarrow \mathfrak{R}$ is said to be continuous at $x = a$ where $a \in S$ if

- $\lim_{x \rightarrow a} f(x)$ exists finitely
- $\lim_{x \rightarrow a} f(x) = f(a)$.

Example 6.3

Let

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

Then

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{x-1} = 2.$$

Also

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{x^2 - 1}{x - 1} = x + 1 = 2.$$

Thus, $\lim_{x \rightarrow 1} f(x) = 2 = f(1)$. Hence f is continuous at $x = 1$.

Definition 6.4 (Pointwise continuity): A function $f : S \rightarrow \mathfrak{R}$ is said to be continuous at $a \in S$ if given any $\varepsilon > 0$, $\exists \delta > 0$ (depending both on ε and the point) such that

$$|f(x) - f(a)| < \varepsilon \text{ whenever } |x - a| < \delta.$$

Example 6.4

- a) The function $f(x) = x^2$ is continuous at every $x = a$.

For given any $\varepsilon > 0$, $a \in \mathbb{R}$ $\delta > 0$

can be found such that

$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

Now,

$$\begin{aligned} |f(x) - f(a)| &= |x^2 - a^2| = |(x - a)(x + a)| \\ &= |x - a||x + a| < \delta|x + a| \end{aligned}$$

Choose $\delta = \frac{\varepsilon}{|x + a|}$. Then whenever $|x - a| < \delta$ we have

$$|x^2 - a^2| = |x - a||x + a| < \frac{\varepsilon}{|x + a|}|x + a| = \varepsilon.$$

Hence $f(x) = x^2$ is continuous at every $x = a$.

- b) Prove that $f(x) = x^2 + 2x + 6$ is continuous at $x = 3$.

Solution

Given $\varepsilon > 0$ we show that $\exists \delta > 0$ such that $|x - 3| < \delta$ implies

$$|f(x) - f(3)| < \varepsilon.$$

Now,

$$\begin{aligned} |f(x) - f(3)| &= |x^2 + 2x + 6 - 21| = |x^2 + 2x - 15| = |(x + 5)(x - 3)| \\ &= |x + 5||x - 3| < \delta|x + 5| \end{aligned}$$

Choose $\delta = \frac{\varepsilon}{|x + 5|}$, then whenever $|x - 3| < \delta$ we have

$$|f(x) - f(3)| < |x + 5||x - 3| < |x + 5|\frac{\varepsilon}{|x + 5|} = \varepsilon.$$

Hence $f(x) = x^2 + 2x + 6$ is continuous at $x = 3$.

Definition 6.5 (Discontinuity of a function): Points where a function fails to be continuous are called points of discontinuity of the function. A function is said to be discontinuous at a point x if it is not continuous at that point.

Example 6.5

Let

$$f(x) = \begin{cases} \frac{x^2 - a^2}{x - a}, & x \neq a \\ 2, & x = a \end{cases}$$

Then $f(x)$ is discontinuous at $x = a$ since

$$\lim_{x \rightarrow a^+} f(x) = 2a = \lim_{x \rightarrow a^-} f(x) = 2a \neq 2 = f(a).$$

Uniform continuity**Recall (Pointwise continuity)**

We said that $f : S \rightarrow \mathbb{R}$ is said to be continuous at $x = a$ if for every $\varepsilon > 0$, $\exists \delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta$, where δ may depend on both x and ε .

That is for different points in S corresponding to pre-arranged $\varepsilon > 0$ we get different values of δ . If we can find a uniform δ that serves for all point of S corresponding to given $\varepsilon > 0$, then we say that f is uniformly continuous on S .

Definition 6.6 (Uniform continuity): A function f defined on S is said to be uniformly continuous on S if for every $\varepsilon > 0$, $\exists \delta > 0$ (depending on ε only) such that

$$|f(x) - f(y)| < \varepsilon \text{ whenever } |x - y| < \delta \text{ for all } x, y \in S.$$

Example 6.6

Prove that $f(x) = 2x$ is uniformly continuous on $\mathbb{R}$.

Proof

Given $\varepsilon > 0$ we want to make $|f(x) - f(y)| < \varepsilon$ by making x sufficiently close to y . That is we must show that for each $\varepsilon > 0$, $\exists \delta > 0$ such that

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

Now,

$$|f(x) - f(y)| = |2x - 2y| = 2|x - y| < 2\delta. \text{ Choose } \delta = \frac{\varepsilon}{2} \text{ depending only on } \varepsilon. \text{ Then}$$

whenever $|x - y| < \delta$ we have

$$|f(x) - f(y)| = 2|x - y| < 2 \cdot \frac{\varepsilon}{2} = \varepsilon$$

Hence f is uniformly continuous on $\mathbb{R}$.

Example 6.7

Prove that $f(x) = x^2$ is uniformly continuous on $[-5, 5]$.

Solution

Now

$$\begin{aligned} |f(x) - f(y)| &= |x^2 - y^2| = |(x - y)(x + y)| \\ &\leq |x - y||x + y| \\ &\leq 10|x - y| < 10\delta \end{aligned}$$

Choose $\delta = \frac{\varepsilon}{10}$ (depending on ε alone). Then whenever $|x - y| < \delta$ we have

$$|f(x) - f(y)| \leq 10|x - y| < 10 \cdot \frac{\varepsilon}{10} = \varepsilon.$$

Thus $f(x) = x^2$ is uniformly continuous on $[-5, 5]$.

Example 6.8

The function $f(x) = \frac{1}{x}$ is uniformly continuous on $[2, \infty)$.

Proof

Given $\varepsilon > 0$ and $x, y \in [2, \infty)$ we have,

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{x} - \frac{1}{y} \right| = \frac{1}{xy} |x - y| \\ &\leq \frac{1}{4} |x - y| < \frac{1}{4} \delta \end{aligned}$$

Choose $\delta = 4\varepsilon$, then whenever $|x - y| < \delta$ we have,

$$|f(x) - f(y)| < \frac{1}{4} |x - y| < \frac{1}{4} \cdot 4\varepsilon = \varepsilon.$$

Negation of the definition of uniform continuity on S

A function $f : S \rightarrow \mathbb{R}$ is not uniformly continuous on S if there exists $\varepsilon > 0$ such that $\forall \delta > 0, \exists x, y \in S$ such that $|x - y| < \delta$ but $|f(x) - f(y)| \geq \varepsilon$.

Example 6.9

The function $f(x) = \frac{1}{x}$ is not uniformly continuous in $(0, 1)$.

Solution (BWOC)

Suppose that f is uniformly continuous in $(0, 1)$ so that $\forall \varepsilon > 0, \exists \delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon \text{ whenever } |x - y| < \delta, \quad \forall x, y \in (0, 1)$$

Let $x = \delta$ and $y = \frac{\delta}{1+\varepsilon}$, where $0 < \delta < 1$.

Then $|x - y| = \left| \delta - \frac{\delta}{1+\varepsilon} \right| = \left(\frac{\varepsilon}{1+\varepsilon} \right) \delta < \delta$.

But

$$\begin{aligned} |f(x) - f(y)| &= \left| \frac{1}{x} - \frac{1}{y} \right| = \left| \frac{1}{\delta} - \frac{1+\varepsilon}{\delta} \right| \\ &= \frac{\varepsilon}{\delta} > \varepsilon \end{aligned}$$

Thus $|f(x) - f(y)| > \varepsilon$ which contradicts the fact that f is uniformly continuous in $(0,1)$.

Example 6.10

Show that $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Solution

Let $\varepsilon = 1$ and $\delta > 0$ be given. Let $x = \frac{1}{\delta}$ and $y = \frac{1}{\delta} + \frac{\delta}{2}$. Then $|x - y| = \frac{\delta}{2} < \delta$. But

$$|f(x) - f(y)| = |x + y||x - y| = \left| \frac{1}{\delta} + \frac{1}{\delta} + \frac{\delta}{2} \right| \frac{\delta}{2} > \frac{2}{\delta} \cdot \frac{\delta}{2} = 1.$$

Thus f is not uniformly continuous on $\mathbb{R}$. (why?) .

Suppose $\varepsilon = 2 > 0$, would it as well work. We must show that given any $\varepsilon > 0$, $\exists x, y \in \mathbb{R}$ such that

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| \geq 2$$

For x , if we let $y = x + \frac{\delta}{2}$

$$\text{Then } |x - y| = \frac{\delta}{2} < \delta.$$

To make

$$|f(x) - f(y)| = |x - y||x + y| < \varepsilon$$

We need to have $|x + y| \geq \frac{2}{\delta}$. This prompts us to choose $\delta = \frac{1}{8}$ (How?).

6.3 Exercise

1. Let $f(x) = \frac{x^2 - 4x - 5}{x - 5}$ for $x \neq 5$. How should $f(x)$ be defined so that it is continuous at?

2. Let $f(x) = x \sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $f(0) = 0$. Show that $f(x)$ is continuous at $x = 0$.
3. Prove that $f(x) = 3x - 5$ is uniformly continuous on $[-1, 1]$.